Marhaba, how may I help you? Effects of Politeness and Culture on Robot Acceptance and Anthropomorphization

Maha Salem
Carnegie Mellon University in Qatar
P.O. Box 24866
Doha, Qatar
MahaSalem@web.de

Micheline Ziadee
Carnegie Mellon University in Qatar
P.O. Box 24866
Doha, Qatar
michelinekz@gmail.com

Majd Sakr
Carnegie Mellon University in Qatar
P.O. Box 24866
Doha, Qatar
msakr@qatar.cmu.edu

ABSTRACT

How do politeness strategies and cultural aspects affect robot acceptance and anthropomorphization across native speakers of English and Arabic? Previous work in cross-cultural HRI studies has mostly focused on Western and East Asian cultures. In contrast, Middle Eastern attitudes and perceptions of robot assistants are a barely researched topic. We investigated culture-specific determinants of robot acceptance and anthropomorphization by conducting a between-subjects study in Qatar. A total of 92 native speakers of either English or Arabic interacted with a receptionist robot in two different interaction tasks. We further manipulated the robot's verbal behavior in experimental sub-groups to explore different politeness strategies. Our results suggest that Arab participants perceived the robot more positively and anthropomorphized it more than English speaking participants. In addition, the use of positive politeness strategies and the change of interaction task had an effect on participants' HRI experience. Our findings complement the existing body of cross-cultural HRI research with a Middle Eastern perspective that will help to inform the design of robots intended for use in cross-cultural, multi-lingual settings.

Keywords

Cross-cultural HRI studies, Communication Style, Natural Language Dialogue, Politeness Theory

1. INTRODUCTION

Interactive service robots are increasingly being employed to provide people with information in public settings, e.g. at reception desks or information booths. As these domains target a diverse range of human interlocutors of different ages and educational backgrounds, one major challenge in designing such robots is the adaptation to different user profiles and expectations. In more international settings such as airports or museums that host visitors from a variety of countries, these challenges take on a whole new dimension by requiring the robot's appearance, non-verbal behaviors and multilingual skills to be appealing and acceptable to users of different cultural backgrounds.

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Most existing cross-cultural studies in human-robot interaction (HRI) investigate salient cultural differences between Western (e.g. America, Europe) and East Asian (e.g. Japan, China, Korea) groups of participants, e.g. [2, 5], typically highlighting major differences regarding user expectations, perceptions and acceptance of robots. For example, religion is believed to influence cultural views and acceptance of robots; Buddhism attributes soul and spirits to both living and non-living creatures, thereby possibly facilitating robots' acceptance in East Asian societies [12]. In contrast, the strong influence of Christianity on Western culture, which holds a clear distinction between objects that do or do not have a soul, may increase the emotional distance between humans and humanlike machines [2].

Only very little research has focused on the study of Middle Eastern attitudes regarding social robots. But since the culture in this region differs substantially from both Western and East Asian cultures, findings and implications from other studies cannot be easily transferred. Moreover, people in the Middle East and other Arabic speaking countries – although commonly labeled as "Arabs" – historically represent very diverse groups of cultural and religious beliefs. In addition, each Arab country (or even region within it) has its own dialect used in everyday communication, while Modern Standard Arabic (MSA) is typically used in formal settings only, e.g. in political speeches or by news presenters. Thus it is difficult to make predictions regarding user expectations and acceptance of robots in these societies. For example, given a strong influence of Islamic culture, which traditionally discourages the depiction of living beings as this is seen as mimicking God by adopting the role of creator, negative attitudes towards humanoid robots may be suspected [15].

In light of these challenges, designing robotic assistants that are intended to serve a broad mass of Arabic speaking nationals is not a straightforward task. In addition, by attempting to target both Arabic and English native speakers in multi-cultural settings as found in the Gulf region, which hosts expatriates from many different countries, the complexity of the endeavor increases. Our work therefore aims to examine cross-cultural differences between Arabic and English native speakers by means of what presumably represents the largest HRI experiment conducted in the Middle East to date. We invited a total of 92 participants from these two cultural groups to interact with a bilingual receptionist robot at Carnegie Mellon University (CMU) in Qatar, in order to investigate factors that may affect robot perception and acceptance of HRI. Our findings will help to inform the design of service robots that can be employed in multi-cultural settings including Arabic speaking audiences.

2. BACKGROUND AND MOTIVATION

Our work draws inspiration from previous research in HRI, anthropology, linguistics and social psychology, and combines them in an interdisciplinary approach. Relevant background information that motivates our work and informed the design of our experiment is summarized in the following.

2.1 Cross-Cultural Differences in HRI

A number of studies have investigated the impact of cultural background on HRI (e.g. [2, 5]), concluding that people from different cultures are likely to respond differently to robots. However, most cross-cultural HRI studies typically focused on Western versus East Asian cultures; in contrast, only little research has been carried out with regard to Middle Eastern attitudes and expectations with regard to robots that are to be integrated into human society.

One of the first – and to date most comprehensive – HRI studies targeting cultural attitudes towards robots in an Arabic speaking country was conducted by Riek et al. [15]. The Ibn Sina humanoid robot, modeled after a famous Islamic philosopher and doctor, was placed in a mall in the UAE where people were invited to interact with it; subsequently, they filled out a questionnaire evaluating their interaction experience and perception of the robot. Although exploratory in nature and mainly comparing ethnic groups from within the region, the study indicates that Arabs generally had positive views on the robot. However, it remains unclear whether this observation was linked to the famous personality that the robot depicted, or whether the respondents' opinions were based on their general concepts of robots. Importantly though, the results showed that Arab participants from the Gulf region (e.g. UAE, Qatar, Saudi Arabia, Yemen) expressed significantly more favorable views with regard to humanoid robots than participants who originate from African regions (e.g. Egypt, Morocco, Tunisia, Libya). These findings highlight potential intra-group differences amplifying the challenge of designing robots that are to interact with Arabs from a variety of different countries.

In a previous study conducted in Qatar with the same robot platform used in the present work, Makatchev et al. [11] investigated how ethnically salient verbal and nonverbal behaviors such as facial appearance can be used to evoke ethnic attributions to a robot character. The results suggested that the robot's perceived ethnicity was more evident for attributions as a native speaker of English rather than a native speaker of Arabic. However, we speculate that this effect may have been caused by the fact that interactions with the robot took place in English only, both for participants of English speaking and Arabic speaking background. Another limitation of the study was the required typed entry of user utterances, which limited participants' perceptions of the robot during interaction. Therefore, our present experiment draws upon the insights from this study and overcomes its limitations by allowing for natural spoken user input and by providing an Arabic speaking version of the robot for Arab participants in addition to the version that interacts with English native speakers using American English.

2.2 Politeness as a Communication Facilitator

Relevant literature in linguistics and cultural anthropology suggests that cultural differences are reflected in different communication styles [14]. Asian and Arabic cultures, for example, are high-context, i.e. people from such cultures are not as verbally direct as people from low-context cultures (e.g. Germans); a substantial part of the delivered message is understood from non-verbal cues and indirect speech, thus decoding the message relies heavily on observing subtle, paralinguistic and context-related cues [20]. In contrast, speakers from low-context cultures tend to code the meaning and intent of a message within the words of the utterance themselves, without relying on paralinguistic aspects to explain the message. As a result, modeling communication styles for a multilingual robot that is intended to address a variety of cultural nationals requires careful consideration.

Claimed to apply universally to dialog across all languages and cultures [3], politeness theory appears like a reasonable starting point to explore and develop cross-culturally acceptable robot behavior. In their seminal work, Brown and Levinson [3] developed their theory around the notion of 'face' which each social actor possesses. As face represents a public image that we seek to protect, any acts that undermine that image are considered face-threatening acts (FTAs). The role of politeness is to mitigate such acts and minimize their threat, which is achieved through four types of strategies employed in communication: *bald on record* (direct strategy, i.e. no attempts to minimize threats to the hearer's face: e.g. "close the window"), *positive politeness* (avoid giving offense by highlighting common ground and friendliness: e.g. "let's close the window, please"), *negative politeness* (avoid giving offense by showing consideration: e.g. "would you mind closing the window?") and *off-record* (indirect strategy: e.g. "it's cold in here"). The choice of strategy depends on factors like social distance and power relations between the interaction partners as well as the extent to which the act is face-threatening.

However, although politeness theory is believed to be universally valid, different cultures may attribute different degrees of face-threat to a potential FTA. In fact, research in the areas of cross-cultural linguistics and pragmatics reveals the existence of cultural differences in peoples' evaluation of and preference for certain politeness strategies. For example, in a study comparing speakers of American English to speakers of Saudi Arabic, Tawalbeh and Al-Oqaily [19] found that speakers of American English preferred the *conventional indirectness* politeness strategy (e.g. by saying "can you close the window, please?") when addressing their inferiors, and only used the direct strategy ("close the window") in cases of close intimacy between speakers and when the speaker's request had a low imposition rate. Regardless of the weight of the imposition, Saudi Arabic speakers, in contrast, used conventional indirectness when addressing superiors and preferred to use the direct strategy when addressing inferiors and friends. Comparing politeness strategies used by speakers of Australian English to those of Palestinian Arabic, Farahat [7] states that members of the first group preferred conventional indirectness while those of the second group opted for "the strategy of apologizing". His findings further reveal that Australian and Palestinian cultures share similar concepts of face – one that is connected with 'respect', 'honor' and 'dignity' – but differ in their understandings of what causes loss and enhancement of face.

In view of these aspects, we assume that the use of politeness strategies can lead to an enhanced HRI experience for users from different cultural groups. But are politeness strategies equally useful in different interaction contexts? For example, when asking a robot for directions, should it be apologetic or direct about the fact that it does not know the answer? And should the robot use the same strategies in a less goal-directed interaction like a small talk conversation? To address these questions, our experimental design incorporates a manipulation of the robot's politeness strategy in two experimental conditions for each cultural group.

2.3 Group Membership as a Predictor of Anthropomorphism

Social psychological research has shown that humans intuitively use cues like a person's ethnicity, age or gender to form impressions of others and categorize them as in-group or out-group members; this, in turn, may result in in-group bias which evokes favorable attitudes, perceptions and behaviors towards members of one's own group [18]. More specifically, Haslam et al. [10] argue that the perception of out-group members as less than human, i.e. *dehumanization*, is an everyday phenomenon, e.g. reflected in abusive language that likens other people to nonhuman entities.

Eyssel and Kuchenbrandt [6] employ such measures established for the study of dehumanization in humans conversely to assess the extent to which a robot is *anthropomorphized*, i.e. viewed as humanlike. In their study, they showed German students identical pictures of a robot, however, half the participants were told that it had been developed in Germany and was called Armin (a German first name). The other participants were told the robot's name was Arman (a typical Turkish name) and that it had been developed at a Turkish university. The students who evaluated the supposedly German-built robot Armin were found to rate it more favorably and anthropomorphized it more than the students who evaluated the supposedly Turkish-built Arman. The authors conclude that humans apply social categorization processes and show a preferential inter-group bias even with regard to a robot: they see it as more human if they believe it shares a common cultural or ethnical background.

These findings led us to predict that the level to which our participants will anthropomorphize the robot will depend on and correlate with their perception of it as an in-group vs. out-group member of their own respective cultural group.

3. METHOD

We conducted an experiment to gain a deeper understanding of how politeness and interaction context might impact and shape human experience and evaluation of HRI, and how these perceptions may vary between participants of two very different cultural backgrounds.

3.1 Hypotheses

Based on findings from related work on cross-cultural studies, anthropomorphism and politeness strategies, we developed four main hypotheses for our experiment:

1. *Effect of condition.* Regardless of the participants' cultural background, manipulation of the robot's politeness strategy will affect
   (a) their perception of the robot and the interaction (subjective assessment of HRI).
   (b) their performance at recalling the route directions presented by the robot (objective assessment of HRI).
2. *Effect of HRI task.* The change of HRI context from goal-oriented interaction (*Task 1*) to open dialogue (*Task 2*) will affect participants' perception of the robot and subjective assessment of HRI.
3. *Effect of cultural background.* Participants' cultural background will affect their perception of the robot and subjective assessment of HRI. In particular, we predict that Arabic participants will have a stronger preference for the polite robot than participants from the group of English native speakers.
4. *Effect of group membership on anthropomorphic inferences.* If one cultural group is found to have a stronger perception of the robot as an in-group member, the same group will be found to make greater anthropomorphic inferences regarding the robot than participants from the other cultural group with weaker perceptions of the robot's in-group membership.

3.2 Experimental Design

We conducted a between-subjects study in which participants interacted with the Hala receptionist robot [17], which consists of a stationary anthropomorphic torso with a screen mounted on a pan-tilt unit (see **Fig. 1a**).

Two language groups were tested: English native speakers and Arabic native speakers. Participants of each respective group interacted with the robot either in English or in Arabic language. For this purpose, the verbal interaction content to be used by the robot during the experiment as well as questionnaires for data collection were translated from English into Arabic by an accredited translator.

For each language group, we further manipulated the *politeness level* of the robot's verbal behaviors in two experimental conditions to measure how this manipulation might impact participants' task performance, perception of the robot and HRI experience in general and with regard to culture-specific differences. Our manipulations were informed by the theoretical framework provided by Brown and Levinson [3], as well as by results from previous experiments using the Hala platform, especially [11]. Specifically, our 'polite' condition was designed to follow the principle of *positive politeness*, while the control condition was inspired by the *bald on record* strategy. To additionally validate their effectiveness, verbal stimuli for the two conditions were evaluated prior to running the experiment by means of two online

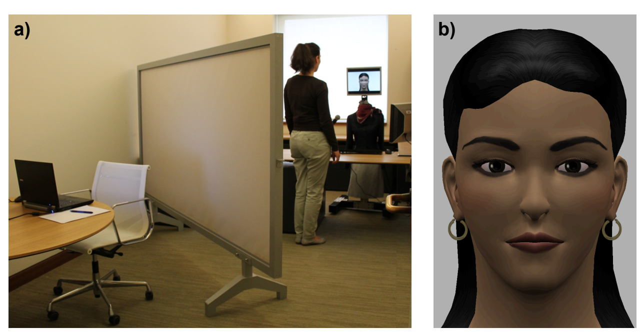

Figure 1: a) Experimental setting; b) the robot's facial expression set to neutral.

questionnaires (one for English, one for Arabic). Each questionnaire comprised the original audio files to be uttered by the robot in the respective language, organized in 32 pairs of utterances which could be replayed with a mouse click. For each language group, 13 independent raters – qualified as native speakers of the corresponding language – were asked to identify the polite version of each of the 32 pairs, resulting in an overall inter-rater agreement of 86.84 % for the English stimulus content and of 87.37 % for the Arabic material.

The robot's face, providing the main channel for the expression of non-verbal behaviors such as emotions, remained identical across conditions and was set to a neutral facial expression as shown in **Fig. 1b**. Visemes were generated to accompany the robot's speech output, for which the Acapela text-to-speech system [1] was used to produce utterances in American English and in Modern Standard Arabic (MSA). The decision to refrain from using gaze and emotional display via facial expressions was motivated by the objective to focus on the verbal interaction level, in order to avoid effects caused by possible interactions between the two modalities.

The experiment comprised two consecutive tasks, each representing a different HRI context as follows.

3.2.1 Direction-giving task

In the first interaction task, participants were given a list with names of three locations in a fictitious building and were instructed to ask the robot for directions to these places. The robot's verbal utterances varied across experimental conditions for dialogue acts of greeting, direction-giving as well as handling failure and disagreement when giving directions. The robot first greeted the participant either with the phrase "Hello, how may I help you?" in the polite condition group or only with the word "Hello" in the control group. Participants were instructed to then ask for directions to Dr. Miller's office, which consisted of five steps (e.g. "turn right", "walk straight until you reach a fountain") and varied across the two conditions based on the use of politeness markers such as "please" or "you will want to turn right" as opposed to "turn right". Since we used participants' ability to recall these directions to measure their task performance, the directions were pretested to be feasible but not trivial to remember. For the second location on the list, participants had to ask the robot for directions to the bookshop, which triggered the robot's failure handling behavior. In the polite condition the robot responded with "Sorry, I don't know where it is, because I am new" in contrast to the response "I have absolutely no idea" in the control condition. Finally, for the third location, participants had to ask for the cafeteria and were instructed to imagine that they misunderstood the robot after being given the directions by asking to clarify if they should "turn left?". This triggered the robot's disagreement handling behavior: in the polite condition, this resulted in the response "Yes, *turn right*", i.e. avoiding explicit disagreement as would be expected in high-context cultures; in contrast, in the control condition the response was "*No*, turn right". The robot's response to the participant's final utterance (typically a variation of "thank you" or "goodbye") did not differ across conditions and was either "you're welcome" or "goodbye" as appropriate.

3.2.2 Chitchat task

For the second interaction, participants were instructed to have an open conversation with the robot, but they were informed that the robot's knowledge was still under development and thus limited. In reality, restricting the robot's knowledge base was an experimental design choice made to ensure a comparable interaction experience across participants. Following a short greeting ("Hello. It's nice to meet you! How are you?" in the polite versus "Hello. How are you?" in the control condition), participants could ask the robot any questions they liked. If the knowledge base did not provide an answer to a question asked, the robot responded by outlining some topics available for discussion (e.g. "Unfortunately, I have no reply to this question. How about talking about Qatar, Carnegie Mellon University, my job or my hobbies, instead?" in the polite condition versus "I have no reply to this question. We can talk about Qatar, Carnegie Mellon University, my job, and my hobbies, instead" in the control condition). As with the first task, the robot's verbal utterances varied across the two conditions based on the use (or lack) of politeness markers such as "please" when asking the participant for something and "I'm very sorry" or "unfortunately" when handling failure to respond.

Depending on the language group to which the respective participant belonged, the robot spoke either in English or Arabic throughout both interaction tasks. We deliberately refrained from randomizing the order of the two interactions tasks, as the unstructured nature of the chitchat task could result in substantial differences in participants' initial perceptions of the robot depending on how the interaction proceeded and what dialogue content was triggered. This impression, in turn, could then affect participants' assessments of the robot after the more structured direction-giving task.

3.3 Experimental Procedure

The experiment was conducted in a closed office room on the CMU campus in Qatar. Using a movable wall (see Fig. 1a), the interaction area was separated from the table at which participants filled out questionnaires. Participants were tested individually in their native language (English or Arabic). They were first greeted by the experimenter outside the room where they received a brief description of the experimental process. After reviewing and signing the consent form, they were asked to fill out a short questionnaire recording their demographic background and previous experience with robots and technology. Participants were then introduced to the first interaction, the *direction-giving task*. Following this task, participants were asked to sit at a laptop provided on the table in the same room and to fill out a questionnaire evaluating the robot and their HRI experience. Upon completion, participants received instructions for the second interaction, the *chitchat task*. After the second interaction, they were asked to fill out another questionnaire on the laptop to document their interaction experience.

To ensure minimal variability in the experimental procedure while allowing for interaction based on natural speech, which is not currently supported by the system, the robot was controlled using a Wizard-Of-Oz technique [9] during the study. The experimenter initiated the robot's verbal interaction behavior from a selection of pre-determined utterances. The ordering of the utterance sequence remained identical for each experimental run within the same condition group for the first task (*direction-giving*), but differed for the second task (*chitchat*) depending on the participants' input. The entire interaction was recorded using a microphone and a voice recorder, while the experimenter listened to and controlled the interaction from an adjacent room.

The second questionnaire was followed by an interview in which the experimenter invited participants to describe and comment on their experience in response to open-ended questions. After the interview, participants were carefully debriefed about the purpose of the experiment and received a tartlet as a thank-you before being dismissed. The total experiment time was approximately 30 minutes, including about 7 to 10 minutes interaction time with the robot.

3.4 Dependent Measures

Dependent variables comprised a number of **subjective measures** based on two questionnaires (one after each interaction task). These were used to tap various dimensions of HRI including participants' perception of the robot, their general HRI experience and their enjoyment of and involvement in the interaction tasks. Five-point Likert scales were used, with high values indicating high agreement with the assessed items. To compute indices for the proposed dependent measures, scores of the included items were averaged after conducting reliability analyses (Cronbach's α).

Politeness We measured perceived politeness of the robot based on two items: 'the robot is polite' as well as the reversely coded ratings of 'the robot is rude' ($\alpha = .742$). This measure served as a manipulation check to verify that our manipulations of the robot's verbal behaviors were effective.

Competency Participants were asked to report how they rated the following competency-related attributes with regard to the robot: 'competent', 'helpful', 'effective', 'cooperative', 'organized' and 'intelligent' ($\alpha = .879$).

Extroversion The degree of extroversion with regard to the robot's personality was assessed using four traits: 'extrovert', 'outgoing', 'sociable', and 'fun-loving' ($\alpha = .809$).

Warmth We measured perceptions of the robot's interpersonal warmth, a core dimension of human social cognition [8], by using the reversely coded ratings of the six items 'cold', 'hardhearted', 'shallow', 'aggressive', 'rude' and 'impatient' ($\alpha = .783$). Since these items were adapted from measures of typically human traits proposed by Haslam et al [10], the warmth index can be used as an indicator of anthropomorphization in HRI (see, for example, [16]).

Psychological Closeness We administered the three items 'How close do you feel to the robot', 'How much do you think you have in common with the robot?', and 'How pleasant was the interaction with the robot for you?' ($\alpha = .812$) to assess participants' degree of psychological closeness to the robot [4]. This index taps perceptions of similarity with the robot and thereby covers further aspects of anthropomorphization as well as HRI acceptance.

After each interaction, participants were asked to specify which nationality they thought the robot represents; this data was collected using open-ended questions. To investigate participants' assessments of the *robot's group membership*, answers were coded in a binary fashion with 0 = robot does not represent a national from the same native language group as the participant's and 1 = robot represents a member of the same native language group as the participant's.

In addition to these subjective measures, assessment of *task performance* based on participants' ability to recall the directions to Dr. Miller's office was used as an **objective measure** (with binary values 0 = incorrect and 1 = correct).

3.5 Participation

A total of 92 participants took part in the experiment, comprising 44 English native speakers (24m, 20f) and 48 native speakers of Arabic (24m, 24f). Participants in the English native speaker group (in the following referred to as *En*) originated from or had spent major periods of their lives in the US, Canada, UK or Australia. The group of Arabic native speakers (further referred to as *Ar*) comprised nationals from Qatar, Egypt, Tunisia, Lebanon, Syria, Jordan, Morocco and UAE. Participants ranged in age from 18 to 70 years ($M = 33.84$ years, $SD = 11.84$) and were recruited at CMU in Qatar. Based on five-point Likert scale ratings (1 = *very little*, 5 = *very much*), they were identified as having negligible experience with robots ($M = 1.96$, $SD = 1.07$) and advanced skills regarding technology and computer use ($M = 4.28$, $SD = 0.91$). Within their respective language group (i.e. English or Arabic), participants were randomly assigned to one of two experimental conditions that manipulated the politeness level of the robot's verbal behaviors, while maintaining gender- and age-balanced distributions.

4. RESULTS

We conducted repeated-measures ANOVA to compare participants' ratings after interaction task 1 and 2, using the between-subjects factors 'culture' (En vs. Ar), 'condition' (polite vs. control) and 'gender' (male vs. female).

Results confirmed that the manipulation of *politeness* level of the robot's verbal utterances had a highly significant effect on participants' ratings of how polite they found the robot, $F(1{,}84) = 18.96$, $p < .001$, $\eta^2 = .18$, with higher ratings from participants in the polite compared to the control condition for both interactions. In addition, the manipulation of the robot's politeness strategy yielded a significant effect of culture, $F(1{,}84) = 9.31$, $p = .003$, $\eta^2 = .10$, with Ar participants generally rating the robot as more polite than En participants (**Fig. 2a**). There was a significant interaction effect between the task and gender $F(1{,}84) = 4.10$, $p = .046$, $\eta^2 = .05$: female participants rated the robot as less polite after the first task than male participants; however, they gave it higher ratings for perceived politeness after the second task compared to their male counterparts whose ratings dropped after the second interaction round (**Fig. 2b**).

Our analysis showed a main effect of interaction task on the robot's perceived *competency*, $F(1{,}84) = 6.73$, $p = .011$, $\eta^2 = .07$: participants of both cultural groups found the robot generally more competent after the first interaction task involving direction-giving than after the second task comprising the chitchat interaction (**Fig. 2c**). Between-groups comparisons yielded a highly significant main effect of participants' cultural background on ratings of the robot's competency, $F(1{,}84) = 14.36$, $p < .001$, $\eta^2 = .15$: Ar participants perceived the robot generally as more competent than En participants (**Fig. 2d**).

There was a highly significant main effect of HRI task on participants' ratings of the robot's level of *extroversion*, $F(1{,}84) = 41.54$, $p < .001$, $\eta^2 = .33$: participants of both cultural groups perceived the robot as more extrovert after the second (chitchat) interaction than after the first (direction-giving) task (**Fig. 2e**). A significant interaction effect on ratings of the robot's perceived level of extroversion was found between participants' cultural background and gender, $F(1{,}84) = 4.79$, $p = .031$, $\eta^2 = .05$: En males rated the

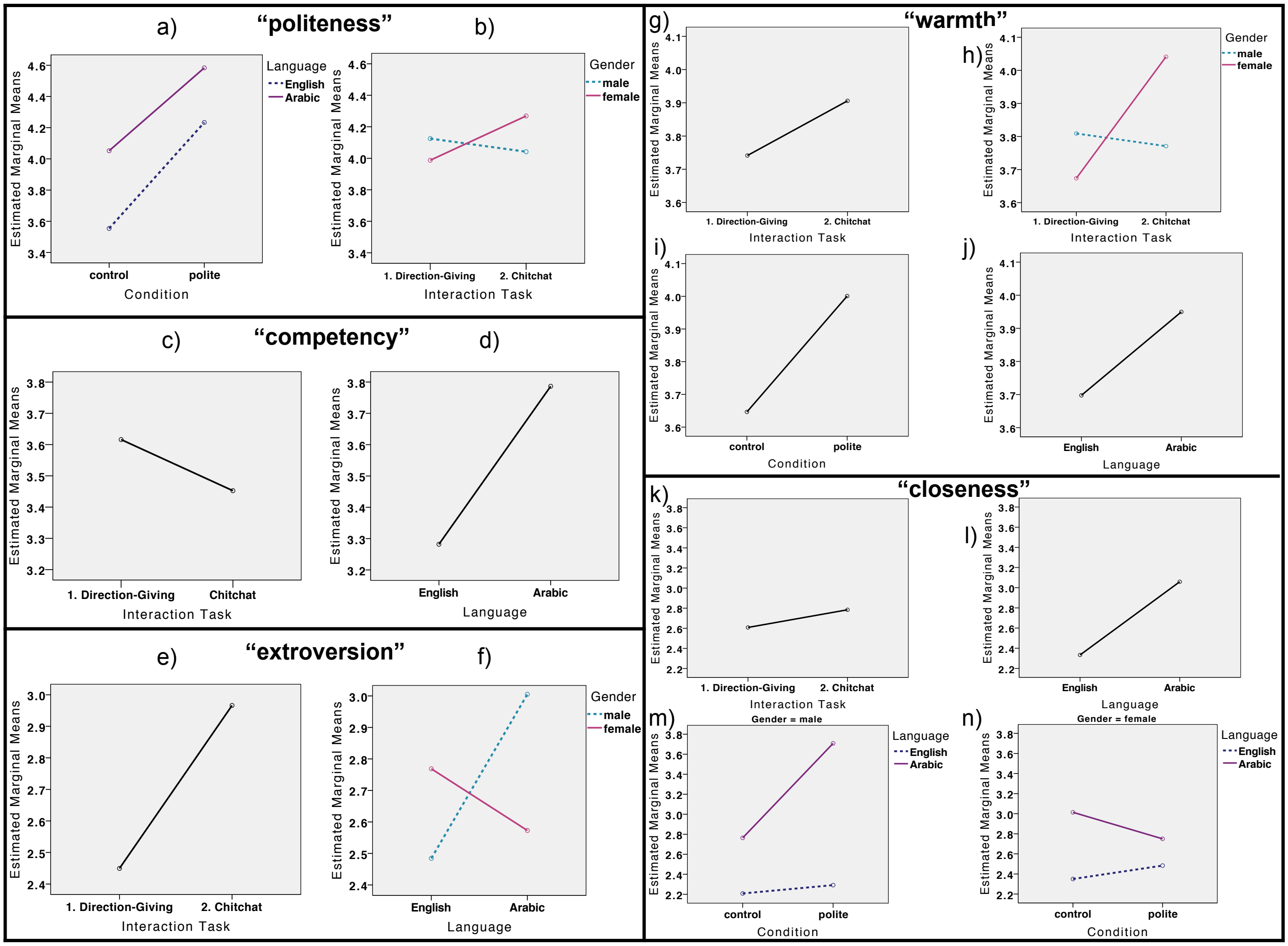

Figure 2: Plots visualizing significant results for the independent measures *politeness* (a&b), *competency* (c&d), *extroversion* (e&f), *warmth* (g–j) and *psychological closeness* (k–n).

robot as less extrovert than Ar males, whereas En females rated the robot as more extrovert than Ar females (**Fig. 2f**).

Our analysis showed a significant main effect of interaction task on participants' ratings of perceived *warmth*, $F(1{,}84) = 6.93$, $p = .01$, $\eta^2 = .08$: participants reported greater perceived warmth after meeting the robot for the second interaction (chitchat) than after the first task involving direction-giving (**Fig. 2g**). We also found a significant interaction effect between HRI task and gender, $F(1{,}84) = 10.52$, $p = .002$, $\eta^2 = .11$: female participants perceived less warmth towards the robot after the first task than male participants, but gave it higher ratings for perceived warmth after the second task compared to their male counterparts whose ratings dropped after the chitchat interaction (**Fig. 2h**). Between-groups comparisons showed a significant main effect of experimental condition on participants' perceptions of warmth towards the robot, $F(1{,}84) = 7.49$, $p = .008$, $\eta^2 = .08$: both En and Ar participants gave higher ratings on the warmth measure when in the 'polite' condition compared to those in the control group (**Fig. 2i**). We also found a marginal main effect (i.e. significant at 10 % level) of participants' cultural background on ratings of perceived warmth, $F(1{,}84) = 3.79$, $p = .055$, $\eta^2 = .04$: regardless of interaction task and condition, Ar perceived greater warmth towards the robot than En participants (**Fig. 2j**).

We found a significant main effect of interaction task on participants' reported experience of *psychological closeness*, $F(1{,}84) = 6.84$, $p = .011$, $\eta^2 = .08$: regardless of the experimental condition, participants of both cultural groups felt closer to the robot after the second interaction task involving the chitchat encounter than after the direction-giving task (**Fig. 2k**). Between-groups comparisons further showed a highly significant main effect of participants' cultural background on ratings of psychological closeness to the robot, $F(1{,}84) = 20.43$, $p < .001$, $\eta^2 = .20$: Ar participants generally reported greater closeness to the robot than En participants (**Fig. 2l**). There was also a marginally significant interaction effect between participants' cultural background, gender and experimental condition $F(1{,}84) = 3.84$, $p = .053$, $\eta^2 = .04$: while Ar males in the 'polite' condition felt closer to the robot than those in the control condition (**Fig. 2m**), Ar females in the control group reversely reported greater closeness to the robot than those in the 'polite' group (**Fig. 2n**). In contrast, both male and female En participants felt closer to the robot when in the 'polite' condition as opposed to those in the control group.

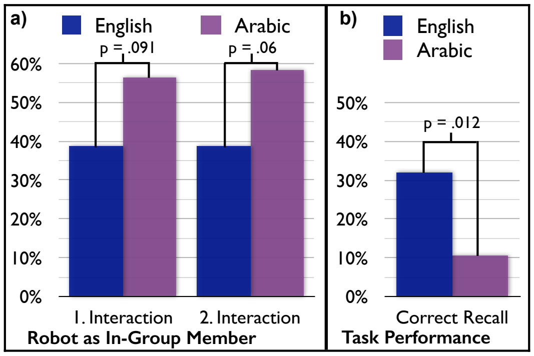

Figure 3: a) Ratings of the robot's in-group membership; b) participants' task performance used as an objective measure

Our analyses further revealed a marginal effect of cultural background on participants' assessment of the *robot's group membership* both after the first interaction, U = 870.00, Z = -1.68, p = .093, and more pronounced after the second round, U = 848.00, Z = -1.88, p = .06: Ar participants were generally more inclined to rate the robot as an in-group member, with 56.3 % after the first and 58.3 % after the second interaction respectively categorizing it as an Arabic native speakers. In contrast, following each interaction round, only 38.6 % of En participants considered the robot to be representing a native English speaker (**Fig. 3a**). No effect of experimental condition or interaction task was found, i.e. neither the change of interaction task nor the robot's politeness strategy affected the ratings of the robot as an in-group member by participants of either cultural group.

Finally, we found no effect of condition on participants' *task performance* when asked to recall the route directions to Dr. Miller's office, U = 1035.00, Z = -0.26, *n.s.* However, there was a significant effect of cultural background on participants' task performance, U = 830.00, Z = -2.52, p = .012: in the En group, 31.8 % of the participants recalled the route directions to Dr. Miller's office correctly, as opposed to only 10.4 % in the Ar group (**Fig. 3b**).

5. DISCUSSION

The results support **Hypothesis 1a** which predicted an effect of experimental condition, i.e. that the manipulation of the robot's politeness strategy will affect participants' subjective assessment of the robot and HRI. Besides the expected differences with regard to ratings of the manipulation check variable *politeness*, the robot's politeness strategy also influenced anthropomorphic inferences made regarding the robot: participants in the polite condition reported significantly greater warmth with regard to the robot than those in the control group. **Hypothesis 1b**, however, was not supported: the use of positive politeness in the robot's verbal instructions did not influence participants' task performance when asked to recall the directions previously given by the robot. Interestingly, an unexpected cultural difference was revealed: Ar participants performed significantly worse at the recall task compared to En participants. One possible explanation is based on cross-cultural linguistics studies suggesting that people differ culturally in the ways in which they provide route directions (e.g. [13]): as the verbal direction-giving content was translated from English into Arabic, the implied strategies might not comply with Arabic cultural standards, thus making it more difficult for Ar participants to follow the instructions. This observation requires further research, while it also emphasizes the importance of cultural and task-related adaptation of robotic assistants.

Hypothesis 2 predicted an effect of interaction task on participants' perception of the robot. Indeed, the results confirm that the robot was anthropomorphized more, with significantly higher ratings on the measures *warmth* and *closeness* as well as on the *extroversion* scale, in the chitchat interaction compared to the direction-giving task. This is not too surprising, as the chitchat task introduced a more personal communication level with the robot which, e.g., told participants about its 'hobbies' and took initiative by asking the user back about their own hobbies. It is similarly not surprising that the robot was perceived as less competent in the chitchat task, as the robot's limited knowledge was challenged by the unstructured nature and potential topical broadness of the open dialog, which easily revealed the robot as less 'flawless' than in the direction-giving task.

According to **Hypothesis 3**, we expected an effect of cultural background, which was confirmed by our results for all dependent measures. Specifically, since related work highlights Arab's preference for a communication strategy of apologizing [7], and the robot was more apologetic in the polite condition when being unable to respond to a question, we predicted that Ar participants would have a more positive perception of the polite robot than En participants. This was reflected by significantly higher Ar ratings for the measure *politeness* and is further supported by significantly higher ratings from Ar compared to En participants on the robot's *competency* scale: despite the general drop of the robot's perceived competency after the chitchat encounter (see Fig. 2c), Arabs seemed more forgiving in their ratings of this measure. One interpretation may be attributed to greater social expectations of politeness in such high-context cultures which typically discourage direct criticism, as which this measure may be perceived. In a way, participants may have even felt that they would be criticizing the developers rather than the robot itself, e.g. one Ar participant reported a first impression as follows: "I felt that the robot is smart and that there are people who worked hard to get to this result." Further supporting the third hypothesis and thus emphasizing cultural differences in HRI, higher ratings from Ar participants on the *warmth* and *closeness* scales suggest that the Ar group anthropomorphized the robot more than the En group. This, in turn, is in line with **Hypothesis 4** which predicted an effect of group membership on anthropomorphic inferences with regard to the robot: as visualized in Fig. 3a, Ar participants were more inclined to categorize the robot as an in-group member than En participants. Although this effect may be partly attributed to the rather Arabic appearance of the robot's face, our results nevertheless confirm the correlation between group membership and anthropomorphism regarding robots, as suggested by previous work [6], even for the extended scope of Arab cultures.

In addition to these hypothesized effects, our results revealed several interaction effects with regard to participants' gender. For example, both for the *politeness* and the *warmth* scale, ratings from female participants' increased notably after the chitchat interaction, while the ratings from male participants dropped. We speculate that females may have been more forgiving regarding the robot's declined competence in the chitchat task, which may possibly be related to the robot's depicted female gender resulting in another in-group effect. Adding another dimension to it, interaction effects between language group and participants' gender on the *extroversion* scale and between condition, language and gender on the *closeness* scale indicate that gender-related differences may play another important role in (cross-cultural) HRI. However, since these effects are outside the main scope of our study, they will require further, more focused, research, e.g. by comparing a male robot with a female version within this cross-cultural setting.

6. CONCLUSION

In this paper, we explored culture-specific variations of robot acceptance and anthropomorphization; our results highlight significant differences in HRI between Arabic and English native speakers, as well as between different interactional contexts (direction-giving vs. chitchat), communication styles (polite vs. direct) and genders.

Importantly, they suggest that Arabs show a more positive attitude towards our robot, including greater anthropomorphization and categorization as an in-group member. However, it is not clear what the exact determinants for these effects are. Among other factors such as the robot's appearance, we speculate that the Ar group's favoring effect may reflect their appreciation of the robot's apparent ability to 'understand' their individual Arabic dialects – something which modern speech recognition software can hardly achieve. To shed light on these questions, future work will include a thorough analysis of our collected qualitative data.

Since our study currently does not provide an exhaustive causal explanation for the observed effects, our findings open up multiple avenues for future work and emphasize the need to further investigate the subtleties of cultural influences on HRI. Such studies, together with our findings, will enable robot designers and programmers to address and exploit the cultural differences when targeting the deployment of multilingual and cross-cultural service robots.

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